

GIANLUIGI VITALE

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EDUCATION

B.Sc. in Computer Engineering — *Universitas Mercatorum, Rome, Italy* Expected Feb. 2027

GPA: 29.03/30 (Italian scale) · 3.79/4.0 (WES-converted)

Thesis: DSProfiler - Static Analysis and Machine Learning for Automated Detection of Data Structure

Performance Anti-patterns in Python

Supervisor: Prof. Rocco Pietrini

Scientific High School Diploma — *Liceo Scientifico "S. Cannizzaro," Palermo* 2014–2019

Full Stack Web Development Bootcamp — *Boolean SRL (700 hours)* Jan.–Jun. 2020

PUBLICATIONS & MANUSCRIPTS

G. Vitale, "DriftBench: Measuring and Predicting Infrastructure Drift in LLM Serving Systems," *Proceedings of MLSys 2026*, Bellevue, WA, May 2026. Oral presentation. DOI: 10.5281/zenodo.19361066. Recorded talk: <https://youtu.be/KnsVj5DdlEo>

G. Vitale, "Decomposing One Professional-Framing Pipeline: Which Components Shift LLM Safety Boundaries?," *NeurIPS 2026 main track*. First-author submission, under review.

G. Vitale, "Natural Language Is Flatland for Machine Intelligence: AI Systems Should Decouple Their Reasoning Substrate from Human Language," *NeurIPS 2026 position paper track*. First-author submission, under review.

RESEARCH INTERESTS

Machine learning systems, LLM serving infrastructure, accelerator portability, TPU kernels and numerical behavior, ML benchmarking, inference reliability, and LLM safety evaluation.

PROFESSIONAL EXPERIENCE

Systems Engineer — *Italian Ministry of Culture* Apr. 2024–Present

Designed and deployed a Retrieval-Augmented Generation system over 10,000 internal administrative documents, delivering source-cited answers to staff queries while minimizing hallucinations. Pioneered the first AI-assisted public consultation interface in an Italian state archive, enabling both employees and citizens to search and cross-reference the full inventory database through natural language, fundamentally transforming how the institution serves its users.

Web Developer — *Informamuse S.R.L.* Oct. 2020–Apr. 2021

Developed websites and web applications using HTML, CSS, JavaScript, PHP, Laravel, Vue.js, and MySQL. Managed projects end-to-end from design through deployment.

ACADEMIC SERVICE

Ethics Reviewer — *NeurIPS 2026* 2026

Reviewed submissions for ethical concerns, potential societal harms, research-integrity risks, and compliance with the conference's ethics guidelines.

Artifact Evaluation Committee Member — SOSP 2026

2026

Evaluated research artifacts for availability, functionality, reproducibility, and consistency with the experimental claims presented in the associated papers.

Artifact Evaluation Committee Member — MLSys 2026

2026

Reviewed four accepted papers covering attention kernel optimization (ThunderKittens/FlashAttention-3), full-stack ML profiling (XProf), GPU memory disaggregation, and distributed inference. Reproduced experimental results on rented GPU clusters (8×H100 via RunPod) and AMD HPC infrastructure (MI350X through the AMD University Program). Recommended the artifact that ultimately received the Distinguished Artifact Award.

RESEARCH EXPERIENCE

- **DriftBench and TPU validation:** Built a measurement and prediction framework spanning 236,985 prompt-response pairs across 105 configurations. The Portability Risk Index generalizes to unseen hardware ($R^2 = 0.909$). Extended the result to TPU v5e/v6e, where predictability held across a new numerical path ($R^2 = 0.72$ - 0.96).
- **DeepSeek V4-Flash on TPU:** Completed a working 32-chip v4-64 bring-up through vLLM/tpu-inference, validating all attention modes with prefill and decode operational. To my knowledge, this is the first TPU bring-up of the model; upstream PRs are drafted.
- **DSA kernel for TPU:** Developed a working implementation that is bit-exact against the reference through 8K tokens and runs through 256K. To my knowledge, it is the first TPU DSA implementation. Work in progress; it is the foundation of the GLM-5.2 port.
- **Bit-level TPU MXU characterization:** Reverse-engineered TPU v5e/v6e numerical behavior and built a CPU simulator that reproduces real Llama-3.1-8B matrix multiplications bit-for-bit on a native pass. Targeting MLSys 2027.
- **Long-context genomics with CNR Italy:** Collaborating on whether long-context DNA models encode linkage disequilibrium, combining ML systems work with genomics expertise at Italy's National Research Council.
- **Structured three-stage framing:** Conducted approximately 31,900 trials across nine LLMs using a factorial ablation of a professional-framing jailbreak, isolating the effects of persona framing, terminology substitution, and moral justification.

TECHNICAL SKILLS

Languages: Python, JavaScript, PHP, SQL, HTML/CSS, Bash

Frameworks: PyTorch, JAX, Pallas, Hugging Face Transformers

ML/Systems: vLLM, tpu-inference, SGLang, TensorRT-LLM, CUDA, Docker

Infrastructure: Google Cloud TPU (v4, v5e, v6e), Linux administration, networking, cloud GPU provisioning (RunPod, AMD HPC/Chameleon Cloud), CI/CD

Research Tools: LaTeX, Git, Jupyter, scipy, statsmodels

COMPUTATIONAL RESOURCES

- Google Cloud TPU Research Cloud (TRC) - active since May 2026. Current 320-chip allocation: 32 on-demand v4 plus 288 spot chips across v4, v5e, and v6e in US and EU zones.
- AMD University Program account (active since March 2026)
- Chameleon Cloud allocation (2,000 SUs)
- RunPod GPU cluster experience (H100, H200, B200, MI300X configurations)

LANGUAGES

Italian (native), **English** (C1 — academic proficiency, scientific publications)

CONTACTS & PROFILES

Google Scholar: <https://scholar.google.com/citations?hl=en&user=x4K6NWUAAAAJ>

ORCID: <https://orcid.org/0009-0007-6740-2695>

OpenReview: https://openreview.net/profile?id=~Gianluigi_Vitale1

Portfolio: <https://gianluigivitale.github.io/portfolio/>

GitHub: <https://github.com/GianluigiVitale>

LinkedIn: <https://www.linkedin.com/in/gianluigi-vitale615>